

ZAPORIZHZHIA, Ukraine

WMO: 346010

Lat: 47.8786N

Lon: 35.0825E

Elev: 112

StdP: 99.99

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-14.6	-20.9	0.6	-17.1	-17.6	0.8	-13.9	10.3	-5.9	9.4	-4.6	3.7	350	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	34.4	20.3	32.8	20.0	30.9	19.4	22.2	29.8	21.4	29.0	20.7	28.1	4.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	14.9	24.8	19.1	14.1	24.0	18.2	13.3	23.4	65.7	29.8	62.9	29.0	60.2	28.3	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	-20.7	36.9	4.2	2.1	-23.7	38.4	-26.1	39.7	-28.5	40.9	-31.6	42.4		
(5)			-20.9	24.0	4.1	1.0	-23.9	24.7	-26.3	25.3	-28.6	25.9	-31.6	26.7		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-3.4	-2.2	3.0	10.6	16.9	20.8	23.2	22.6	16.9	9.7	3.6	-1.2
(7)		DBStd	10.53	5.72	5.64	4.36	4.05	4.23	3.52	3.19	3.71	4.22	4.61	4.70	5.35
(8)		HDD10.0	1629	416	341	218	41	3	0	0	0	2	62	199	347
(9)		HDD18.3	3449	674	574	474	233	79	16	2	6	76	268	443	606
(10)		CDD10.0	1670	0	0	2	59	216	323	409	392	209	54	6	0
(11)		CDD18.3	447	0	0	0	1	34	89	152	139	32	2	0	0
(12)		CDH23.3	4662	0	0	0	16	352	871	1568	1539	302	15	0	0
(13)		CDH26.7	1857	0	0	0	3	99	307	662	696	88	2	0	0
(14)	Wind	WSAvg	3.6	3.9	4.1	4.3	3.8	3.3	3.3	3.2	3.4	3.4	3.4	3.7	4.0
(15)	Precipitation	PrecAvg	507	40	35	40	39	46	63	45	37	42	35	42	42
(16)		PrecMax	775	110	79	75	84	121	157	122	156	155	69	125	96
(17)		PrecMin	375	10	10	9	1	2	22	10	4	2	3	3	6
(18)		PrecStd	92	22	19	19	26	28	32	34	35	36	18	29	21
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	8.2	11.0	17.6	25.2	32.0	33.9	36.5	37.5	32.2	25.1	17.0	11.2	
(20)		0.4% MCWB	7.1	8.5	10.1	14.9	18.9	20.6	20.9	20.4	19.1	15.8	13.4	9.9	
(21)		2% DB	6.2	8.0	14.5	22.2	29.2	31.9	34.2	35.1	29.1	21.8	14.1	9.0	
(22)		2% MCWB	5.5	6.6	8.9	13.2	17.5	19.7	20.8	20.0	17.7	14.8	12.0	8.3	
(23)		5% DB	4.9	6.1	12.0	20.2	27.2	30.1	32.4	33.0	26.8	19.1	12.0	7.2	
(24)		5% MCWB	4.3	4.7	7.6	12.3	17.0	19.3	20.5	19.6	16.5	14.2	10.6	6.6	
(25)		10% DB	3.0	4.2	9.5	18.1	25.0	28.2	30.6	30.8	24.5	16.7	10.0	5.8	
(26)		10% MCWB	2.5	3.3	6.3	11.1	16.3	18.8	20.0	19.2	15.8	13.0	9.0	5.2	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	7.4	9.1	11.3	15.7	20.7	22.7	23.7	22.8	20.7	17.6	14.1	10.4	
(28)		0.4% MCDB	8.1	10.5	16.0	22.7	27.6	29.3	30.8	31.6	28.3	22.2	16.0	11.1	
(29)		2% WB	5.7	6.8	9.6	14.3	19.2	21.5	22.5	21.7	19.2	16.1	12.4	8.2	
(30)		2% MCDB	6.2	7.7	13.3	20.3	26.0	28.1	30.0	30.0	25.5	19.7	14.0	8.7	
(31)		5% WB	4.1	4.9	8.0	13.1	18.1	20.6	21.7	20.9	18.0	14.8	10.8	6.6	
(32)		5% MCDB	4.6	5.7	10.7	18.5	24.6	27.1	29.3	29.3	23.6	18.1	11.8	7.1	
(33)		10% WB	2.5	3.4	6.6	12.1	17.1	19.7	21.0	20.1	16.9	13.4	9.0	5.1	
(34)		10% MCDB	2.9	4.2	9.1	17.0	23.6	26.0	28.4	27.8	22.4	16.3	9.8	5.6	
(35)	Mean Daily Temperature Range	MDBR	5.3	6.2	8.0	10.6	11.3	11.0	11.6	12.1	10.9	8.9	6.1	5.0	
(36)		5% DB MCDBR	5.3	7.5	12.3	13.9	13.8	13.3	13.8	14.6	13.9	12.1	8.5	5.9	
(37)		5% WB MCWBR	4.7	6.1	7.7	6.9	6.4	5.3	5.0	5.3	6.2	7.0	6.2	5.3	
(38)		5% WB MCDBR	5.1	6.8	10.7	12.2	12.3	11.5	12.1	12.6	11.5	10.7	7.4	5.7	
(39)		5% WB MCWBR	4.7	5.9	7.4	6.6	6.2	5.4	5.0	5.5	5.9	6.9	5.8	5.3	
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.360	0.389	0.449	0.416	0.416	0.440	0.438	0.412	0.374	0.353	0.334	
(41)		taud	2.230	2.184	2.216	2.082	2.270	2.293	2.232	2.226	2.269	2.354	2.324	2.282	
(42)		Ebn at Noon	713	776	818	800	848	849	821	803	787	760	688	662	
(43)		Edh at Noon	80	105	119	150	129	127	133	129	112	88	74	69	
(44)	All-Sky Solar Radiation	RadAvg	0.97	1.83	2.99	4.60	5.92	6.29	6.26	5.64	3.94	2.38	1.10	0.72	
(45)		RadStd	0.17	0.31	0.37	0.41	0.49	0.51	0.35	0.33	0.43	0.34	0.17	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.77	N/A	N/A	+1.26	N/A	N/A	N/A	-146	+116
(47) Regional (9 neighbors)		+0.78	N/A	N/A	+1.14	N/A	N/A	-139	-198	+131

Nomenclature: See separate page